



Lab 5: Secure Digital Transmissions

Introduction to Secure Digital Transmissions

In modern computer networks, the security of data transmissions is crucial. As users interact with websites, they frequently send and receive sensitive information, such as login credentials, payment details, and personal data. If this information is transmitted insecurely, attackers can intercept and misuse it, leading to data breaches, identity theft, and other cybersecurity threats.

Two widely used protocols for transmitting data over the web are:

- **HTTP (Hypertext Transfer Protocol):** The standard protocol used for retrieving web pages and communicating with web servers. However, HTTP transmits data in plaintext, meaning anyone who intercepts the traffic can easily read its contents.
- **HTTPS (Hypertext Transfer Protocol Secure):** An extension of HTTP that encrypts data using **SSL/TLS (Secure Sockets Layer/Transport Layer Security)**. This encryption ensures that data remains confidential and cannot be easily intercepted or altered.

This lab will guide you through capturing and analyzing HTTP and HTTPS traffic using **Wireshark**, a popular network protocol analyzer. By submitting form data over HTTP and HTTPS, you will observe the critical differences between plaintext and encrypted transmissions.

Capturing HTTP Traffic using Wireshark

To analyze HTTP traffic, we will use a simple HTML form provided in Google Classroom. This form contains only a **username** and **password** field and submits data using the HTTP protocol. Since HTTP transmits data in plaintext, we will use Wireshark to capture the network traffic and examine how easily sensitive information can be intercepted.

Start by opening **Wireshark** and selecting the appropriate network interface, either **Wi-Fi** or **Ethernet**, depending on how your computer is connected to the internet. Double click on the given options to begin the packet capture. Once started you can stop by clicking the red button next to the blue fin, or restart by clicking on the **Start Capture** button (the blue fin button). Once Wireshark is recording network traffic, open the HTML form in your web browser and enter test credentials for the **username** and **password** fields. Click the **Submit** button to send the form data over the network.

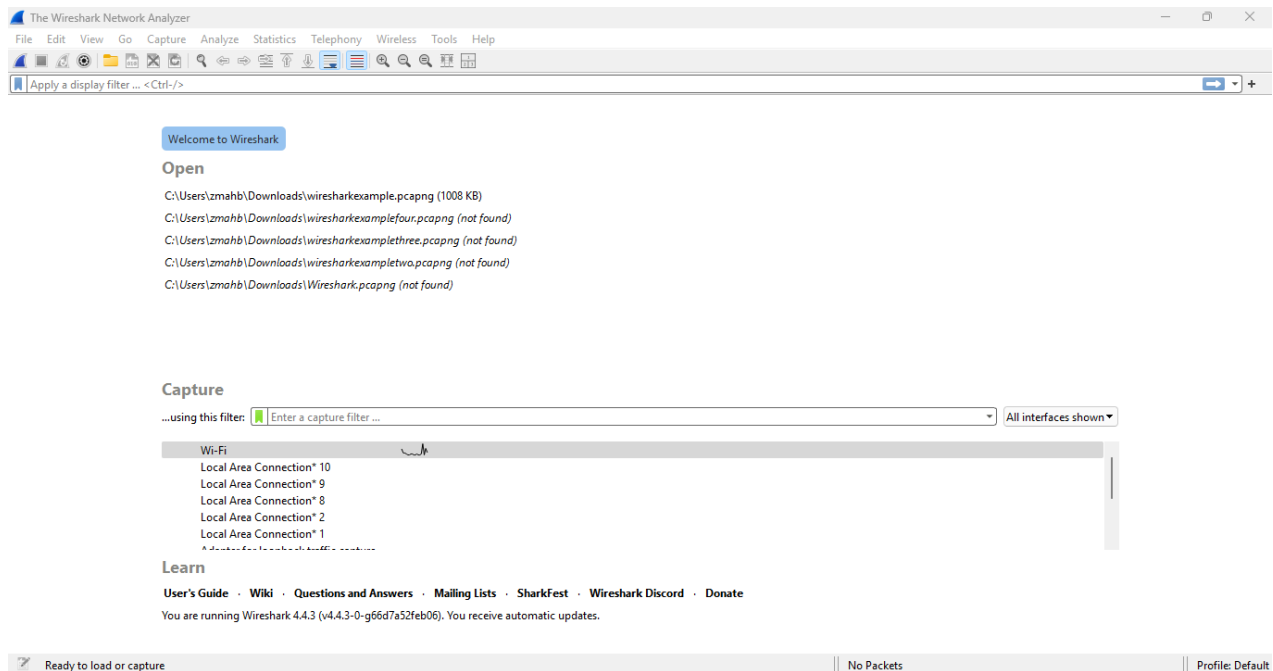
After submitting the form, stop the packet capture in Wireshark. To filter out irrelevant traffic, enter *http* in the filter bar and press **Enter**. This will display only HTTP packets. Look for a **POST request** in the **Info** column, as this is the request that carries the form data. Click on the **POST request** packet to examine its details. In the **Packet Details** pane, expand the **Hypertext Transfer Protocol (HTTP)** section, followed by the **HTML Form URL Encoded**



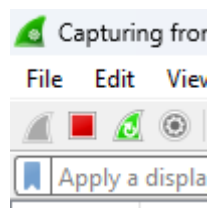
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section. Here, you will be able to see the **username** and **password** in plaintext, exactly as they were entered in the form.

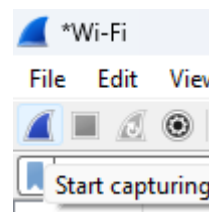
This confirms the fundamental security flaw of HTTP: all transmitted data is unencrypted, meaning that anyone monitoring the network can intercept and read sensitive information such as login credentials. You can also observe some screenshots related to the steps described above, in the sections below.



Selecting Wi-Fi interface for capturing packets.



Stop button



Start button



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HTTP Login Test for Wireshark

D:/CNET/httpexample/test_form.html

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Test Login Form (HTTP)

Username:

Password:

Submit (HTTP POST)

Instructions for Wireshark Capture:

1. Open Wireshark and start capturing on your network interface
2. Use filter: host httpbin.org and http
3. Fill out this form with test credentials
4. Click submit
5. Look for the HTTP POST packet in Wireshark
6. Right-click on the packet and select "Follow > HTTP Stream"

Accessing the website with the HTML form

HTTP Login Test for Wireshark

D:/CNET/httpexample/test_form.html

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Entering insecure credentials



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```
httpbin.org/post
Not secure httpbin.org/post
Pretty-print
{
  "args": {},
  "data": "",
  "files": {},
  "form": {
    "password": "1234",
    "username": "test1"
  },
  "headers": {
    "Accept": "text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7",
    "Accept-Encoding": "gzip, deflate",
    "Accept-Language": "en-US,en;q=0.9",
    "Cache-Control": "max-age=0",
    "Content-Length": "28",
    "Content-Type": "application/x-www-form-urlencoded",
    "Host": "httpbin.org",
    "Origin": "null",
    "Upgrade-Insecure-Requests": "1",
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36",
    "X-Amzn-Trace-Id": "Root=1-67b07d50-4e8b38cb77eb2a951fe1be2e"
  },
  "json": null,
  "origin": "154.192.8.52",
  "url": "http://httpbin.org/post"
}
```

Response page after insecure credentials have been entered

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
458	22.173114	192.168.18.17	3.214.119.249	HTTP	622	POST /post HTTP/1.1 (application/x-www-form-urlencoded)
466	22.414310	3.214.119.249	192.168.18.17	HTTP	922	HTTP/1.1 200 OK, JSON (application/json)

> Frame 458: 622 bytes on wire (4976 bits), 622 bytes captured (4976 bits) on interface \Device\NPF...
> Ethernet II, Src: Intel_ff:6d:92 (18:56:80:ff:6d:92), Dst: HuaweiTechno_f7:a7:5a (6c:44:2a:f7:7f:9c)
> Internet Protocol Version 4, Src: 192.168.18.17, Dst: 3.214.119.249
> Transmission Control Protocol, Src Port: 50668, Dst Port: 80, Seq: 1, Ack: 1, Len: 568
> Hypertext Transfer Protocol
> HTML Form URL Encoded: application/x-www-form-urlencoded

0000 6c 44 2a f7 a7 5a 18 56 80 ff 6d 92 08 00 45 00 ID*..Z.V..m...E
0010 02 60 69 fe 40 00 80 06 40 11 c0 a8 12 11 03 d6 .i@...@.....
0020 7f f9 c5 e4 00 5b d4 47 ea ab c6 2e ba c3 5b 18 w...P.G...P
0030 00 ff ad 9c 00 00 5b 4f 53 54 20 2f 70 6f 73 74PO ST /post
0040 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a HTTP/1.1..Host:
0050 20 68 74 74 70 62 69 6e 2e 6f 72 67 0d 0a 43 6f httpbin.org..Co
0060 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 nnection : keep-a
0070 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 live..Content-Le
0080 6e 67 74 68 3a 20 32 38 0d 0a 43 61 63 68 65 2d ngth: 28 ..Cache-
0090 43 6f 6e 74 72 6f 6c 3a 20 6d 61 78 2d 61 67 65 Control: max-age
00a0 3d 30 0d 0a 4f 72 69 67 69 6e 3a 20 6e 75 6c 6c =0..Origin: null
00b0 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 ..Content-Type:
00c0 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 2d 77 77 applicat ion/x-ww
00d0 77 2d 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f 64 65 w-form-u rlenCode
00e0 64 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 d..Upgra de-Insec
00f0 75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d ure-Requ ests: 1
0100 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a ..User-Ag ent: Moz
0110 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 illa/5.0 (window
0120 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 s NT 10. 0; Win64

Hypertext Transfer Protocol: Protocol

Packets: 801 · Displayed: 2 (0.2%) · Dropped: 0 (0.0%)

Profile: Default

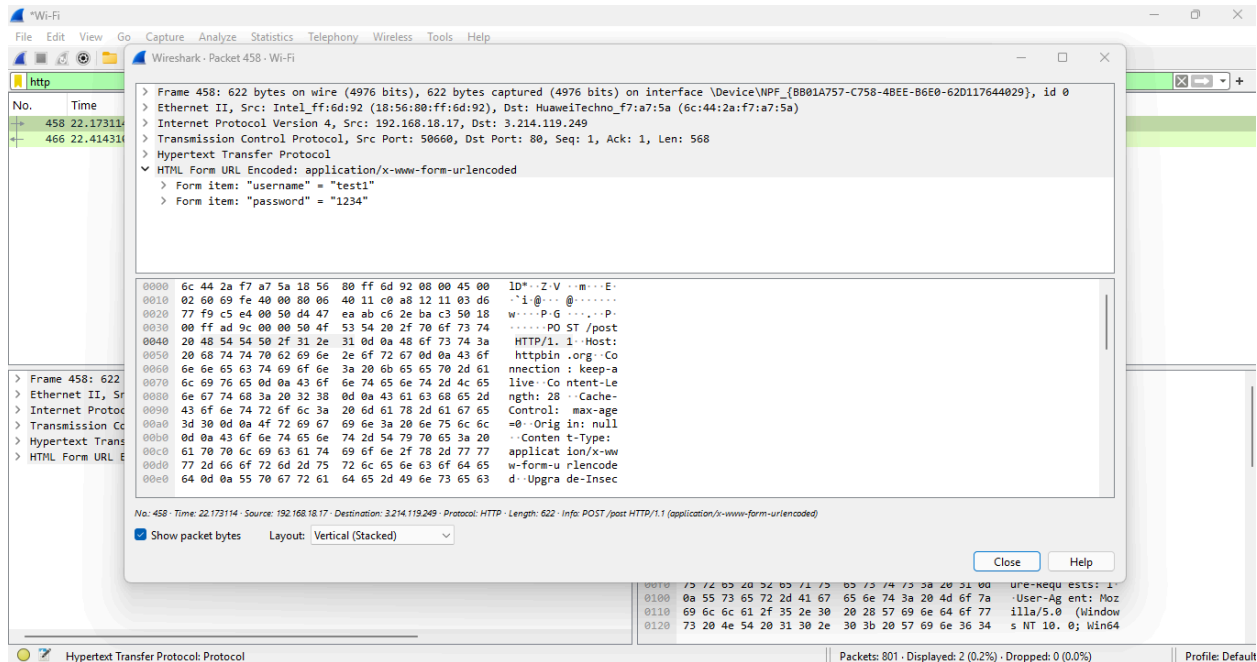
Rain showers
Next Thursday

4:41 PM
15-Feb-25

Filtering packets to show HTTP traffic



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Opening relevant packet to see details sent through HTTP request

Capturing HTTPS Traffic using Wireshark

To analyze HTTPS traffic, we will visit **Wikipedia** and search for the term "**cheese**", then browse through different pages on the website. Unlike HTTP, HTTPS encrypts all transmitted data using SSL/TLS, making it unreadable even when intercepted. Our goal is to capture HTTPS packets in Wireshark and observe how the encryption prevents us from seeing the actual data being transmitted.

Begin by opening **Wireshark** and selecting the appropriate network interface. Click on the **Start Capture** button to begin recording network traffic. With Wireshark running in the background, open a web browser and navigate to <https://www.wikipedia.org/>. In the Wikipedia search bar, type "**cheese**" and press **Enter**. Browse through various cheese-related articles for a minute to generate sufficient network traffic.

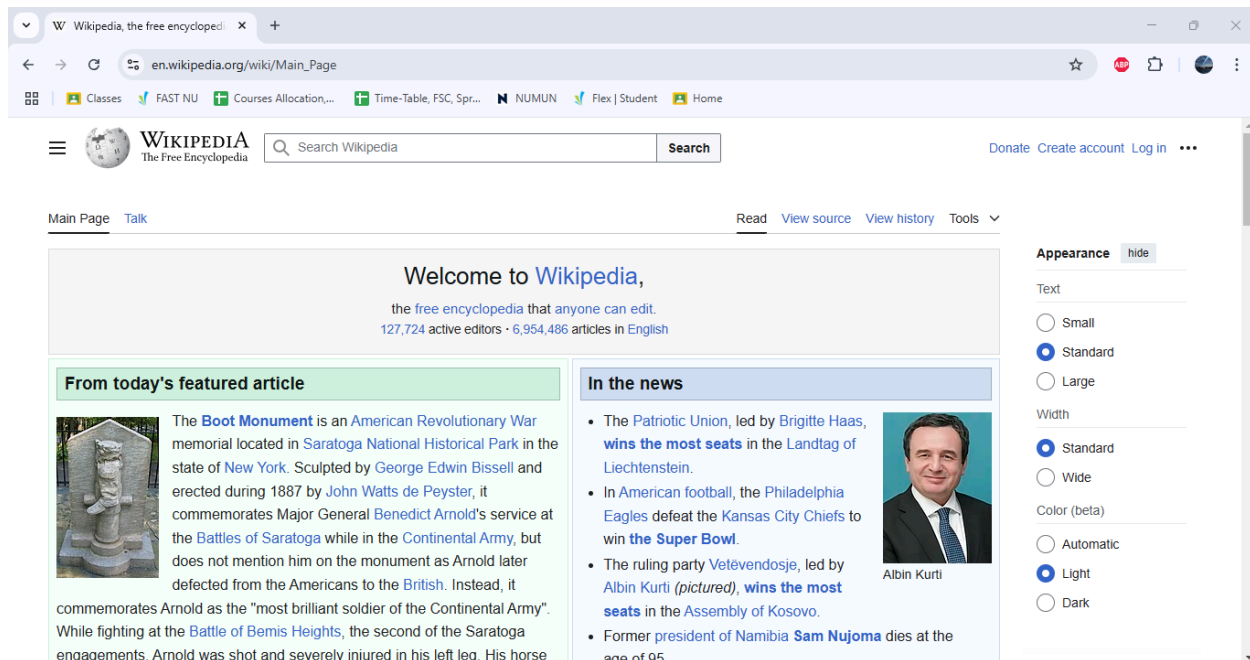
Once you have interacted with the website, stop the packet capture in Wireshark. To isolate HTTPS traffic, apply a filter by entering either `tcp.port == 443` or `ssl` in the filter bar and pressing **Enter**. Unlike HTTP, you will not see clearly readable text in the captured packets. Instead, locate a packet labeled **Application Data** under the **TLSv1.2** or **TLSv1.3** protocol. Clicking on



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this packet will reveal an **Encrypted Application Data** section in the **Packet Details** pane, indicating that the actual content of the transmission is securely encrypted.

This confirms that HTTPS successfully protects sensitive information by encrypting it before transmission. Unlike HTTP, where data is sent in plaintext, HTTPS ensures that even if network traffic is intercepted, the actual content remains hidden from unauthorized parties. This encryption is what makes HTTPS the preferred choice for secure digital communications, safeguarding user privacy and protecting sensitive data from cyber threats. You can also observe some screenshots related to the steps described above, in the sections below.



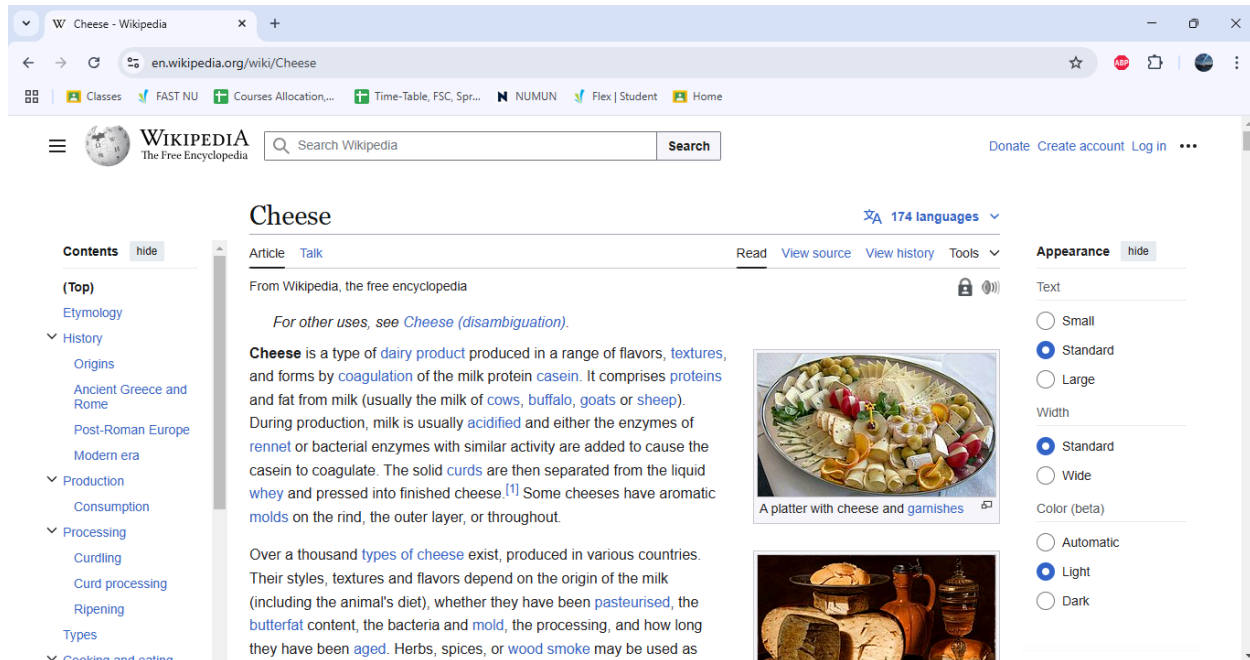
Going to Wikipedia while Wireshark runs in the background



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Results after searching “cheese”



Browsing through different cheese related pages.



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Wireshark capture showing TCP traffic on port 443. The packet list shows a sequence of TCP segments from 192.168.18.17 to 103.102.166.224. The packet details pane shows the selected packet (No. 33) as a Transmission Control Protocol segment with Src Port: 50692, Dst Port: 443, Seq: 1, Ack: 1, Len: 0. The packet bytes pane shows the raw data of the segment.

Applying tcp.port == 443 (the port used for HTTPS traffic) to see HTTPS traffic

Wireshark capture showing HTTPS traffic on port 443. The packet list shows a sequence of TCP segments from 192.168.18.17 to 103.102.166.224. The packet details pane shows the selected packet (No. 33) as a Transport Layer Security (TLS) record layer. The packet bytes pane shows the raw data of the segment, including the TLS record structure.

Observing encrypted application data



Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO3: Evaluate networking technologies and architectures in terms of performance, scalability, interoperability, and security within diverse communication environments.

Before starting, ensure that a notepad or sticky note containing your name, roll number, section, and lab number is clearly visible. All screenshots must include this note, otherwise, they will not be considered.

Task 1: Analyzing HTTP Traffic (10 marks)

Using the code given in the google classroom, you will generate HTTP traffic and show insecurely sent data clearly displayed on Wireshark. For this, download the code from the GCR and, using any IDE, add a field for **age** in the form and add your **roll number and name** in place of `<h1> </h1>`. Then you will follow these steps, **recording each step by taking screenshots of it and attaching them to a pdf file/word file:**

Start the Wireshark Capture

1. Open **Wireshark**.
2. Select the **network interface** that your computer uses for internet access (e.g., Wi-Fi or Ethernet).
3. Click **Start Capture**.

Submit the Form

4. Open **test_form.html** in a web browser.
5. Enter the following details in the form:
 - **Username:** Your roll number
 - **Password:** Any random password (e.g; 1234)
 - **Age:** Any number
6. Click **Submit**.

Filter and Analyze HTTP Packets

7. Stop the Wireshark capture.
8. In the **Wireshark Filter Bar**, type `http` and press **Enter** to isolate HTTP traffic.
9. Locate the **POST** request packet that contains the submitted form data.



10. Click on the packet and expand the **HTML Form URL Encoded** section in the lower panel.
11. Take a **screenshot** showing the captured username, password, and age in plaintext.

Also, **answer** the following question (in 1-2 lines):

Q1. Why is HTTP considered insecure for transmitting sensitive data? Explain the risks of plaintext transmission and how attackers can exploit it.

Task 2: Analyzing HTTPS Traffic (10 marks)

By navigating through different pages on Wikipedia, you will generate HTTPS traffic that you will then capture and analyze to see how encryption works. For this, you will go to <https://www.wikipedia.org/> and browse through several pages related to tomatoes. The steps you will follow, **recording each step by taking screenshots of it and attaching them to a pdf file/word file**, are:

Start a Wireshark Capture

1. Open **Wireshark** and start a new packet capture session (details also given in task 1).

Visit Wikipedia and look for a page related to Tomatoes

2. Open your web browser and navigate to [Wikipedia's Tomato page](#).
3. Browse the page for up to 1 minute to generate network traffic.

Filtering and Analyzing HTTPS Packets

4. Stop the Wireshark capture.
5. In the **Wireshark Filter Bar**, type `tcp.port == 443` or `ssl` and press **Enter** to isolate HTTPS traffic.
6. Locate a packet labeled **Application Data** (this contains encrypted content).
7. Take a **screenshot** of the packet details showing that the data is encrypted.

Also, **answer** the following questions (in 1-2 lines each):

Q1. What differences do you notice between HTTP and HTTPS traffic? Compare the packet details for HTTP vs. HTTPS and note how the data is secured.

Q2. Why is HTTPS considered more secure than HTTP? Explain the role of SSL/TLS encryption in preventing data interception and tampering



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Submission Guidelines (5 Marks):

- Submit a single **PDF/Word** document containing all tasks to the GCR **AND THE HTML PAGE. (DO NOT SUBMIT THE WIRESHARK FILES)**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1**. Rename the HTML file as **rollnumber_name.html**.
- The usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Demo:

- If student cannot answer at all -5 marks.
- If student can answer somewhat -2 marks.
- Correct answers results in no deductions.

Sample screenshot:

